

# ABHINAV KAUSHIK

## Lead Applied Scientist, Document Understanding

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## SUMMARY

Senior AI/ML Engineer with 8+ years delivering applied machine learning, analytics and production AI systems across Healthcare, Telecom and AIOps. Works across Python, SQL, PySpark, scikit-learn, LightGBM, LLM applications and ML evaluation, from stakeholder requirements through deployment. Delivers measurable outcomes with stakeholders. Alongside this, I am building ThinkTree.AI, a non-profit AI learning platform used by NGO teachers to support students with ADHD.

## PROFESSIONAL EXPERIENCE

### Senior AI Engineer | United Health Group | Bengaluru, India Nov 2025 – Present

- Deployed a production lab-extraction agent with Claude Opus 4.7 and Amazon Textract, integrated with LIMS, handling about 1,800 lab documents.
- Engineered a production multi-agent analytics chatbot in LangGraph for AARP medical plans, using PySpark analysis and multi-turn reasoning, cutting manual analysis by 98%.
- Built automated LLM evaluation and observability pipelines with tracing and failure detection, reducing human-in-the-loop review by 43%.
- Developed a RAG-based Project Discovery Assistant in LangGraph combining semantic retrieval with LLM reasoning, reaching a 90% match rate.

### Senior Machine Learning Engineer | Viavi Solutions | Chennai, India Sep 2024 – Nov 2025

- Developed an internal RAG-based Project Discovery Assistant in LangGraph combining semantic retrieval with LLM reasoning, reaching a 90% match rate against existing organizational solutions.
- Built an event-correlation engine clustering alarms with Sentence-BERT and DBSCAN to trace the cascading effect of network alarm floods back to the root event, improving semantic alarm classification by 73.5%.
- Prototyped 15+ AI applications for AIOps and GeoPerformance clients from client requirements, with 7 moving into production or paid client work.

### Data Scientist | Viavi Solutions | Bengaluru, India Nov 2021 – Sep 2024

- Deployed a production SLA breach prediction system with LightGBM, FastAPI and MLflow, using GPT-3.5 for real-time root-cause analysis and remediation recommendations per breach, reducing downtime by 67%.
- Shipped anomaly detection and demand forecasting on 15+ KPIs across 500+ telecom network devices using Isolation Forest and Holt-Winters, enabling earlier detection and proactive network planning.
- Delivered technical POCs and consulting for 15+ clients across 3+ AIOps and ML platforms, contributing to an 11% increase in POC and mentoring revenue.

### Data Analyst | Optum | Gurugram, India Jun 2021 – Nov 2021

- Implemented a reranker for an existing recommendation system serving individual and child plan recommendations, optimizing renewal suggestions and increasing conversion by 2%.
- Developed a Python and Flask parser that automated extraction and transformation of raw data into structured datasets, feeding downstream analytics and reporting.

### Data Engineer | Nippon Data Systems | Gurugram, India Nov 2019 – May 2021

- Engineered SQL and PySpark data pipelines automating BAU medical-audit reporting on large healthcare datasets, cutting manual reporting effort by 95%.
- Wrote complex SQL queries with multi-table joins, CTEs and window functions, delivering ad-hoc reports and analyses for multiple stakeholder teams.
- Built real-time log observability for the COVID QRT team with Elasticsearch and Kibana, improving traceability efficiency by 80%.

### Associate Consultant | Saksoft | Gurugram, India Dec 2017 – Nov 2019

- Developed an XGBoost propensity model predicting personal-loan applications within 7 days of contact, enabling targeted outreach and customer prioritization.
- Built customer dashboards for financial and product insights with QlikSense, SQL and SAP HANA, increasing user engagement by 40%.

## PROJECTS

### Think Tree – ML Study Platform

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

### RFP Multi-Agent Proposal Automation Platform

- Built a multi-agent RFP platform where an LLM supervisor orchestrates intent detection, retrieval, proposal generation, document matching and market research, with RAG over historical proposals.

#### Financial Fraud Detection – Mar 2025

- Built a fraud detection and reasoning workflow on the JPMorgan Synthetic Dataset, pairing LightGBM scoring with LLM-generated explanations for interpretable financial-risk analysis.

#### Restaurant Feedback Analyzer – Aspect-Based Sentiment Analysis

- Built an NLP pipeline to extract review aspects and determine sentiment at the aspect level, enabling granular insights into customer experience.

## EDUCATION

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|--------------------------------------------------------------------------------------|---------------------|
| <b>Master of Science, Machine Learning and AI   Liverpool John Moores University</b> | Jan 2021 – Jan 2023 |
| <b>Bachelor of Technology, Information Technology   Jamia Hamdard</b>                | Aug 2013 – Aug 2017 |

## TECHNICAL SKILLS

**AI / GenAI:** Generative AI, LLM, RAG, Agentic RAG, Agentic AI, LangGraph, LangChain, Claude, OpenAI, Embeddings, Cross-Encoder, Fine-Tuning, Hugging Face Transformers, NLP, Tool Orchestration, Amazon Textract

**Machine Learning:** PyTorch, Scikit-learn, XGBoost, LightGBM, Random Forest, Isolation Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Regression, Deep Learning

**Evals / Observability:** LLM Evals, LLM Evaluators, Agent Tracing, Observability, Elasticsearch, Kibana, Log Analysis

**Programming / Data:** Python, SQL, PySpark, Pandas, NumPy, FastAPI, Flask, Hive, Spark SQL, REST API, Pydantic, Unix

**MLOps / Cloud:** MLflow, Docker, Kubernetes, Airflow, DVC, Terraform, Jenkins, Bamboo, Bitbucket, GCP, Azure, Vercel

**Data Platforms:** Databricks, Databricks SQL, Apache Spark, Supabase

**Databases:** TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB

**Search:** Apache Solr, Lucene Indexing, Information Retrieval

**BI:** Power BI, QlikSense, SAP HANA

## LANGUAGES

English: Native

German: A1 – Pursuing

## ADDITIONAL INFORMATION

- Network Intern, Airport Authority of India
- Hadoop Intern, Ducat Technologies